

Sravya Aravapalli

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EDUCATION

Carnegie Mellon University

May 2028

B.S Computer Science, Minor in Machine Learning, GPA: 3.5

Pittsburgh, PA

Courses: Machine Learning, Functional Programming, Probability, Distributed Systems, Generative AI

Organizations: Carnegie Mellon Racing, Society of Women Engineers, Alpha Chi Omega: Kappa Nu

TECHNICAL SKILLS

Languages: Python, Java, C++, C, SQL, JavaScript, TypeScript, Go, C#, HTML/CSS

Frameworks: React, Next.js, Node.js, Express, Flask, Spring Boot, PyTorch, pandas, scikit-learn

Tools: Git, Linux/Unix, Docker, Kubernetes, AWS, Jira, PostgreSQL, MySQL, MongoDB, Snowflake

EXPERIENCE

Axon

May 2026 – Aug. 2026

Software Engineering Intern

Boston, MA

- Developed C/C++ firmware for an embedded device to offload IMU sensor data over LTE, implementing data serialization, buffering, and MQTT-based transmission for an end-to-end data-collection pipeline
- Designed concurrent collection and transmission workflows using RTOS multithreading under strict embedded resource constraints; flashed, debugged, and tested firmware directly on target hardware
- Built Python analysis tooling to identify patterns and detect events across 10+ hours of time-series IMU data, achieving 90% event-detection accuracy against human-labeled ground truth
- Delivered the pipeline as the team's primary data-collection tool, enabling 50+ hours of field data used for model training and validation

Carnegie Mellon University

Aug. 2026 – Present

Teaching Assistant

Pittsburgh, PA

- Leading recitations and office hours, assisting in developing and grading assignments and exams, and supporting review sessions for 15-151 Mathematical Foundations for Computer Science
- Teaching proof techniques including induction, combinatorics, and graph theory and reinforcing discrete mathematics foundations for students in a 300+ student course

RESEARCH

CMU Reliable Autonomous Systems Lab

Jan. 2026 – May 2026

Undergraduate Engineering Researcher

Pittsburgh, PA

- Worked under Professor Reid Simmons on interactive robotics system design and development.
- Led the migration of Gamebot's C++ Interaction Manager to Python for an autonomous Scrabble-playing robot used in human-computer interaction research, redesigned its rule-based priority action pipeline to improve scalability and maintainability.

University of Minnesota

June 2025 – Aug. 2025

Undergraduate Researcher ML/AI

Minneapolis, MN

- Selected into 'AI for Earth Summer School', focused on applying AI and machine learning to solve challenges in domains such as climate change, agriculture, and sustainability
- Developed an LSTM based deep learning model to predict NEE, GPP, and RECO Net Ecosystem Exchange (NEE), Gross Primary Productivity (GPP), and Ecosystem Respiration (Reco) metrics, in order to assess carbon dioxide fluxes and forecast ecosystem sustainability at several global sites
- Achieved an RMSE of 0.53 for GPP prediction, corresponding to around 5% of its daily flux range

PROJECTS

Astro Sandbox | *TypeScript, Next.js, React, Three.js, PostgreSQL*

Aug. 2025 – Nov. 2025

- Developed a real-time orbital mechanics simulator with interactive 3D Three.js visualization, Web Worker-based physics computation, and Prisma ORM/PostgreSQL backend APIs